

Ruyi Ding

Email: ruyiding@lsu.edu

Phone1: (+1) 404-271-7158 Phone2: (+86) 178-168-78071

Personal website: rollinding.github.io



ACADEMIC APPOINTMENTS

Louisiana State University, Baton Rouge, USA

Aug. 2025 - Present

Assistant Professor at the Division of Electrical and Computer Engineering

Northeastern University, Boston, USA

Jul. 2020 - Apr. 2025

Graduate Research Assistant

Georgia Institute of Technology, Atlanta, USA

Aug. 2019 - May 2020

Graduate Teaching Assistant

EDUCATION

Northeastern University, Boston, USA

Jul. 2020 - Apr. 2025

Ph.D. in Computer Engineering

Advisor: Prof. Yungsi Fei

Georgia Institute of Technology, Atlanta, USA

Aug. 2018 - May 2020

M.S. in Electrical and Computer Engineering

Advisor: Prof. Yao Xie

Zhejiang University, China

Sept. 2014 - Jul. 2018

B.S. in Information Science & Electronic Engineering

Advisor: Prof. Fan Zhang

RESEARCH INTERESTS

My research centers on Secure AI Infra — securing modern AI systems end-to-end — with three primary thrusts:

- Securing models and data against extraction, probing, poisoning, and misuse;
- Hardware security and confidential computing — side-channel analysis and trusted execution environments for trustworthy AI;
- Security for agentic AI — protecting LLM and Mixture-of-Experts deployments and agent platforms;

These efforts advance one mission: building trustworthy AI infrastructure — from silicon and accelerators to foundation models and autonomous agents. My approach emphasizes both theoretical analysis of modern architectures and practical system implementations.

AWARDS

- **DAC Outstanding TPC Award** DAC 2026, July 2026.
- **DAC Young Fellow** DAC 2025, Apr 2025.
- **Northeastern Outstanding PhD Student Research Award** Mar 2025.
- **Northeastern PhD Network Travel Award** Feb 2025.
- **Internet Society Fellowship at NDSS 2025** NDSS 2025, Jan 2025.
- **Distinguished Paper Award at ASIACCS 2023.** ASIACCS 2023, July 2023.
- **Best Paper Awards Nomination at DATE 2022** DATE 2022, March 2022.
- **Graduate Student External Award.** June 2024.

PUBLICATIONS

- [1] Xiaocheng Zou, Tiancheng Zheng, Xiaolin Xu, **Ruyi Ding**. *Capacity Overflow: A Blind Spot for Backdoor Attacks in Vision MoE*. European Conference on Computer Vision 2026 (ECCV 2026).
- [2] Tianhong Xu, Saion Roy, **Ruyi Ding**, Aidong Adam Ding, Yungsi Fei. *SLAC: Access-Driven CPU-to-GPU Side-channel Attacks via System-Level Cache on Apple Silicon*. ACM Conference on Computer and Communications Security 2026 (CCS 2026).
- [3] Zihao Xin, Xin Zhang, Zhe Liu, Qianmei Wu, Qingni Shen, **Ruyi Ding**, Yixin Jiang, Wenqian Xu, Zhihong Liang, Lu Zhou, Fan Zhang. *Shooting Falcon with One Fault: A Practical Rowhammer-based Fault Attack against Falcon*. Design Automation Conference 2026 (DAC 2026).

- [4] **Ruyi Ding***, Tianhong Xu*, Xinyi Shen, Aidong Adam Ding, Yunsi Fei. *MoEcho: Exploiting Side-Channel Attacks to Compromise User Privacy in Mixture-of-Experts LLMs*. ACM Conference on Computer and Communications Security 2025 (CCS 2025)
- [5] **Ruyi Ding**, Cheng Gongye, Davis Ranney, Aidong Adam Ding, Yunsi Fei. *MACPruning: Dynamic Operation Pruning to Mitigate Side-Channel DNN Model Extraction*. IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2025).
- [6] **Ruyi Ding***, Tianhong Xu*, Aidong Adam Ding, Yunsi Fei. *Graph in the Vault: Protecting Edge GNN Inference with Trusted Execution Environment*. Design Automation Conference 2025. (DAC 2025)
- [7] **Ruyi Ding**, Tong Zhou, Lili Su, Aidong Adam Ding, Xiaolin Xu, Yunsi Fei, *Probe-Me-Not: Protecting Pre-trained Encoders from Malicious Probing*. The Network and Distributed System Security Symposium (NDSS 2025).
- [8] Shijin Duan*, **Ruyi Ding***, Jiaxin He, Aidong Adam Ding, Yunsi Fei, Xiaolin Xu, *GraphCroc: Cross-Correlation Autoencoder for Graph Structural Reconstruction*. The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024).
- [9] **Ruyi Ding**, Lili Su, Aidong Adam Ding, Yunsi Fei, *Non-transferable Pruning*. In European Conference on Computer Vision (pp. 375-393). Springer, Cham. (ECCV 2024)
- [10] **Ruyi Ding***, Shijin Duan*, Xiaolin Xu, Yunsi Fei *VertexSerum: Poisoning Graph Neural Networks for Link Inference*. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 4532-4541). (ICCV 2023)
- [11] **Ruyi Ding**, Cheng Gongye, Siyue Wang, Aidong Adam Ding, Yunsi Fei, *EMShepherd: Detecting Adversarial Samples via Side-channel Leakage*. In Proceedings of the 2023 ACM Asia Conference on Computer and Communications Security. (ASIACCS 2023) **Distinguished Paper Award**.
- [12] **Ruyi Ding**, Ziyue Zhang, Xiang Zhang, Cheng Gongye, Yunsi Fei, Aidong Adam Ding *A cross-platform cache timing attack framework via deep learning*. In 2022 Design, Automation & Test in Europe Conference & Exhibition (DATE 2022). IEEE. **Best Paper Award Nomination**.
- [13] Yize Li, Pu Zhao, **Ruyi Ding**, Tong Zhou, Yunsi Fei, Xiaolin Xu, Xue Lin. *Neural architecture search for adversarial robustness via learnable pruning*. Frontiers in High Performance Computing, 2: 1301384.
- [14] Xiang Zhang, Ziyue Zhang, **Ruyi Ding**, Cheng Gongye, Aidong Adam Ding, Yunsi Fei. *Ran \$ Net: An Anti-Ransomware Methodology based on Cache Monitoring and Deep Learning*. In Proceedings of the Great Lakes Symposium on VLSI 2022, pp. 487-492. 2022.
- [15] Shixiang Zhu, **Ruyi Ding**, Minghe Zhang, Pascal Van Hentenryck, Yao Xie *Spatio-temporal point processes with attention for traffic congestion event modeling*. IEEE Transactions on Intelligent Transportation Systems 23.7 (2021): 7298-7309.
- [16] Shixiang Zhu, Minghe Zhang, **Ruyi Ding**, Yao Xie *Deep Fourier Kernel for Self-Attentive Point Processes*. In International conference on artificial intelligence and statistics, pp. 856-864. PMLR, 2021.
- [17] Fan Zhang, Xiaoxuan Lou, Xinjie Zhao, Shivam Bhasin, Wei He, **Ruyi Ding**, Samiya Qureshi, Kui Ren. *Persistent fault analysis on block ciphers*. IACR Transactions on Cryptographic Hardware and Embedded Systems (2018): 150-172.

The authors are ordered by contribution and (*) indicates that authors equally contributed

INVITED TALKS & PRESENTATIONS

Invited Talks

- **Wuhan University**, School of Cyber Science and Engineering Dec. 2025
Topic: *Side-Channels in AI Systems: Spear or Shield?*
- **Westlake University**, Westlake Youth Forum Nov. 2025
Topic: *Towards Robust and Secure AI System: Bridging Hardware Security for Trustworthy Intelligence*

Conference Presentations

- **ACM Conference on Computer and Communications Security (CCS)**, Taipei Oct. 2025
Paper: *MoEcho: Exploiting Side-Channel Attacks to Compromise User Privacy in Mixture-of-Experts LLMs*
- **IEEE International Symposium on Hardware Oriented Security and Trust (HOST)**, San Jose, CA May 2025

Paper: MACPruning: Dynamic Operation Pruning to Mitigate Side-Channel DNN Model Extraction

- **Network and Distributed System Security Symposium (NDSS)**, San Diego, CA Feb. 2025

Paper: Probe-Me-Not: Protecting Pre-trained Encoders from Malicious Probing

TEACHING & MENTORING EXPERIENCE

Louisiana State University, Baton Rouge, LA

Instructor of Record

- **EE 7700: Introduction to Hardware Security** (Graduate Level) Spring 2026
 - Developing advanced curriculum on side-channel analysis, logic locking, and hardware Trojans.
 - Integrating research-oriented projects to bridge classroom learning with real-world security challenges.
- **EE 4745: Neural Computing** Fall 2025
 - Taught fundamental theories of deep learning, including CNNs, RNNs, and backpropagation.
 - Designed final projects focusing on model deployment and optimization.

Research Advising & Mentoring

- **Current Group:** 2 Ph.D. Student (Co-advisor) and 1 M.S. Student (from UCLA).

PROFESSIONAL SERVICE

Editorial Board

- **Youth Editorial Board Member**, *Digital Twins and Applications* Dec. 2025 – Dec. 2027

Technical Program Committee (TPC) Member

- Design Automation Conference (**DAC**) 2026
- IEEE International Symposium on Hardware Oriented Security and Trust (**HOST**) 2026
- IEEE/ACM International Conference on Computer-Aided Design (**ICCAD**) 2025, 2026
- IACR Transactions on Cryptographic Hardware and Embedded Systems (T-CHES) 2026

Journal Reviewer

- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2026

Conference Reviewer

- Annual Conference on Neural Information Processing Systems (NeurIPS) 2026
- Association for the Advancement of Artificial Intelligence (AAAI) 2026

Conference External Reviewer

- Network and Distributed System Security Symposium (**NDSS**) 2026
- ACM Conference on Computer and Communications Security (**CCS**) 2025

UNIVERSITY SERVICE

Louisiana State University

- **Member**, Faculty Search Committee 2025 – Present
 - Evaluating candidates for tenure-track positions in Electrical and Computer Engineering.